Rohith Kalisetti

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SUMMARY

Embedded firmware engineer with experience developing C/C++ software across FreeRTOS, embedded Linux, and bare-metal environments. Background spans connected-device firmware, hardware-aware software development, embedded networking, automated validation, and GPU programming, with a focus on solving problems across firmware, hardware, and system boundaries.

TECHNICAL SKILLS/TOOLS

Languages: C, C++, Python, Bash, Assembly
Embedded Systems: FreeRTOS, ESP32, MSP430, ATmega32, ARM Cortex-M, DMA, interrupts, timers, UART, SPI, I2C, CAN
Embedded Linux / Network: Linux, OpenWrt, POSIX, Wi-Fi, BLE, TCP/IP, UDP, DHCP, DNS, MQTT, hostapd, wpa_supplicant
Debug / Validation: GDB, Valgrind, ASAN/UBSAN, Ceedling, CMock, tcpdump, Wireshark, oscilloscope, logic analyzer
CI / Build / Tooling: Git, GitHub Actions, Docker, CMake, Make, GHIDRA, QEMU, Jira
GPU / Parallel: CUDA, Pthreads, NVIDIA Nsight, PyTorch, TensorFlow

WORK EXPERIENCE

Firmware Engineer Intern | Mill Industries | San Bruno, CA ***Nov 2025 - Feb 2026***

- Built a C/C++ FreeRTOS simulation layer that decoupled firmware logic from ESP32 hardware, enabling task, timer, delay, and concurrency behavior to be exercised reproducibly on Linux x86 systems.
- Designed an OpenWrt-based fault-injection platform that recreated Wi-Fi, DHCP, DNS, routing, port-blocking, and cloud-connectivity failures, turning intermittent device issues into repeatable engineering test cases.
- Modernized firmware test infrastructure by migrating Ceedling 0.31 to 1.x, rebuilding mocks and platform abstractions, and resolving preprocessing, dependency, include-path, and build-system incompatibilities across firmware modules.
- Integrated the simulated firmware and validation stack into Dockerised QEMU and GitHub Actions workflows with ASAN, UBSAN, Valgrind, coverage analysis, packet captures, and device logs, enabling failures to be detected and diagnosed before hardware testing.

Graduate Teaching Assistant | George Mason University | Fairfax, VA ***Aug 2024 - Oct 2025***

- Supported embedded systems labs involving MSP430 and ARM Cortex-M firmware, interrupts, timers, DMA, UART, SPI, and I²C, helping students diagnose hardware and software faults using oscilloscopes, logic analyzers, and serial traces.
- Guided C, Assembly, Pthreads, and CUDA assignments covering parallel programming, GPU memory hierarchy, kernel execution, thread divergence, and performance profiling with NVIDIA Nsight.

Graduate Research Assistant | George Mason University | Fairfax, VA ***Aug 2024 - May 2025***

- Contributed to the Hardware Debloat research project presented through the DAC Young Fellows programme, building a QEMU/DECAF-based dynamic feature-management system across Windows XP, Windows 7, and Lubuntu guest images.
- Worked within a team that developed four DECAF/QEMU plugins to dynamically control network, Q-device, block-device, and context-switching features, supporting attack-surface reduction, resource efficiency, and full-system benchmarking.
- Developed a Python/GHIDRA static-analysis pipeline that reconstructed control-flow graphs across 40+ firmware binaries, handling indirect calls, computed jumps, recursion, and call over-approximation; correlated static results with live DECAF execution traces.

Senior Quality Analyst | Apple *via* Cognizant | India ***Feb 2022 - Jul 2023***

- Diagnosed and resolved 1000+ geospatial data defects in Apple Maps' production pipeline using Linux-based log analysis, improving routing accuracy and POI integrity at global scale; automated daily validation workflows with Python and shell scripting.

Risk Investigations Specialist | Amazon | India ***Jun 2020 - Oct 2021***

- Investigated 500+ accounts daily using fraud detection platforms, surfacing false-positive trends that informed detection-model refinements; documented findings in Jira and Confluence for cross-team SOP updates.

Technical Support Engineer | Dell Technologies | India ***Sep 2019 - Feb 2020***

- Triaged 40+ daily hardware/firmware escalations using BIOS-level diagnostics and Linux system log analysis, logging structured root-cause reports in Jira that accelerated L3 engineering resolution.

PROJECTS

GPU-Accelerated Image Processing & Parallel Computing *C, CUDA, Pthreads, NVIDIA Nsight*

- Developed multithreaded CPU and CUDA implementations for edge detection, bitmap flipping, image rotation, Gaussian filtering, convolution, and color transformations.
- Optimised CPU and GPU performance using Pthreads, workload partitioning, shared and constant memory, coalesced memory access, precomputation, and tuned thread-block configurations.
- Benchmarked implementations across multiple image sizes, achieving 4-8x speedups while analyzing latency, occupancy, throughput, cache behavior, and PCIe transfer overhead.

Gesture-Controlled Smart Wheelchair with Obstacle Detection *Embedded C, ATmega32, UART/SPI, IR, MEMS*

- Wrote bare-metal firmware on an ATmega32, translating MEMS accelerometer gestures into real-time motor control via interrupt-driven UART/SPI loops; tuned ISR timing to keep actuation latency stable under varying input rates.
- Implemented sensor fusion between the accelerometer and IR proximity sensors, overriding motion commands to halt within a safe stopping threshold for reliable obstacle avoidance.

EDUCATION

M.S., Computer Engineering | GPA: 3.90/4.0 | *George Mason University* **May 2025**